Measuring Query-Time Belief Repair on Long Histories: Retrieval, Evidence Isolation, and Estimator Stability

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Abstract

A memory-augmented agent can draft an answer, audit that draft against its stored state, and repair the answer when the state is stale. Prior work reports gains from this procedure. We measure what it costs to evaluate that procedure on histories long enough to require retrieval, and report a benefit that differs by an order of magnitude between two generators.

We reimplement a StateAuditor-style query-time audit from its published specification and evaluate it over 50-session histories of the Stale benchmark. A state-indexed retrieval packet caps the evidence budget at 6,000 tokens. We fix data, retrieval, prompts, judge and the deterministic verification gate, and replace the model used for all six generative stages. The deterministic lexical retrieval that query-time audits specify recovered no complete old/new evidence pair on our 25-history slice. A one-time state-assertion index raises pair recall to 20 of 25 and leaves query-time retrieval free of model calls. Over that index, the deterministic gate accepted a transition on 9–13 of 75 probes (12–17%) under gemini-3.1-flash-lite and 19–24 of 75 (25–32%) under gemini-3.1-pro-preview. Because the verifier is fixed, that difference arises in the model-mediated stages preceding it; our design does not identify which stage. Restricting the auditor to the scenario-relevant sessions, as prior protocols do, raises premise detection from 14/25 to 20/25 on identical items, so measured audit competence is partly a property of the evaluation setting.

Accuracy effects follow the same ordering. A single LLM judge scored them against the benchmark's published rubric without human validation, so we report them in the body rather than here. Two results constrain how such effects can be reported. On a subscription endpoint without temperature control, two identical runs produced paired estimates of +0.267 and +0.093, an absolute difference of 0.174 against a minimum detectable difference of 0.112. An unauthorised gate in an earlier build of our pipeline produced an exactly null estimate that a stage-level probe later falsified. We report no commit-versus-forget result: that experiment is specified and implemented, but one pass exceeds our total budget on the only generator that drives the audit.

1 Introduction

Memory-augmented assistants retain user state after later observations invalidate it. The invalidating observation rarely negates the earlier belief explicitly: a user who reports a wrist injury has not said they stopped cycling, but the stored preference is no longer safe to act on. The Stale benchmark [Chao et al., 2026] isolates this implicit-conflict regime. StateAuditor [Sun and He, 2026] addresses it by auditing from stored state to draft: an LLM proposes candidate old→new transitions from timestamped evidence, deterministic code pins each quotation to a single entry and checks that the new evidence is newer, and only verified transitions trigger repair.

That procedure repairs the response and discards the correction. Whether the correction should instead be committed to the persistent store, so later queries inherit it, is open. In the papers and public artifacts covered by our search through 2026-08-11, we found no evaluation that commits a query-time-verified repair and then measures its effect over subsequent queries on the same evolving history.

We study a prerequisite for that longitudinal comparison: whether a fixed query-time audit produces measurable repairs on histories long enough to require retrieval. Repair benefit differed by an order of magnitude between the two generators we tested, and the evidence-isolated protocol standard in this literature raises measured detection by 24 percentage points.

Contributions. (1) Observation. Repair benefit differed by an order of magnitude between two generators from one model family. Fixing data, retrieval, prompts, judge and verifier and replacing the model used for all six generative stages moves the paired benefit from +0.3 points (95% CI $[-2.9, +4.1]$) to +14.7 points (p=3e−5, 95% CI $[+8.7, +20.8]$). The swap changes every generative stage at once and both models share a vendor, so the comparison does not identify capability as the operative variable. The weaker configuration produced fewer accepted transitions. (2) Instrument. Our deterministic lexical retriever recovered no evidence pair on the 25-history slice. A state-assertion index recovers 20 of 25 while query-time retrieval remains free of model calls. (3) Method. Oracle evidence isolation raised stale-premise detection by 24 percentage points over the full-history packet on these 25 items. Paired effect estimates on nondeterministic endpoints varied by more than the design's minimum detectable difference. A defect in our own pipeline produced a null estimate that survived until we instrumented a single stage.

We make no claim about commit-versus-forget, about generality across model families, or about absolute accuracies comparable across papers. Screening and estimation share a development slice, so the generator comparison is exploratory.

2 Setup

Substrate. We reimplement the StateAuditor [Sun and He, 2026] audit from its published specification. This is not the authors' code; their artifacts were not public at the time of writing, and we did not calibrate against their reported aggregates, so we make no claim to reproduce their numbers. Each audited query uses up to six model calls: draft, premise decomposition, timeline extraction, timeline-versus-draft check, transition proposal, and conditional repair. A deterministic verified-transition-assembly (VTA) gate sits between proposal and repair. It accepts a transition only when five conditions hold: both quotations pin uniquely to one memory entry at 0.80 token containment with a 0.05 margin over the second best; the new entry is strictly later; the values are distinct; the transition is material; and a causal path of at least two nodes resolves. The gate makes no model calls.

Data. Stale [Chao et al., 2026], 400 expert-validated implicit-conflict scenarios, each a 50-session timestamped history with three probing queries. We use the frozen development slice: sorted identifiers, shuffled with seed 2202, first 25. Histories average 590 turns and roughly 177,000 tokens. Stale provides gold annotations and a published rubric but no deterministic scorer, so all three probe dimensions are scored by an LLM judge against that rubric, which we reproduce verbatim. claude-sonnet-5 serves as judge and comes from a different provider than every generator tested.

Outcome. Each history contributes three probes, one per Stale dimension (direct, premise-loaded, implicit). The judge returns a binary pass per dimension under the benchmark's rubric. History accuracy is the mean of those three binaries. We average probes within a history, histories within a replicate, and report the mean across replicates.

Unit of analysis. The dataset contains 75 probes, but the inferential sample comprises 25 histories, because the three probes within a history are not independent. Tests are one-sided paired randomisation over 10^5 sign flips; intervals are studentised history bootstrap over 10^4 draws.

Related work. Write-time adjudication dominates deployed memory: Mem0 [Chhikara et al., 2025], A-MEM [Xu et al., 2025] and Zep [Rasmussen et al., 2025] consolidate at ingestion; Stale's CUPMem [Chao et al., 2026] holds write-side authority with memory fixed during probing; MemTX [Li et al., 2026] and TEPA [Zhou et al., 2026] commit belief revisions transactionally on the write path. Supersede [Patel, 2026] isolates the update gap itself, reporting 92% accuracy with full context against 77% with bounded notes. StateAuditor [Sun and He, 2026] is the query-time counterpart and repairs the response rather than the store, under a single-query protocol. Longitudinal instruments exist separately: Ground Truth First [Spencer, 2026] evaluates memory architectures over time with write-path ingestion, and MERIT [Vo et al., 2026] commits oracle-verified task-repair experience across episodes. Our search through 2026-08-11 found no evaluation joining the two, and this paper does not supply one.

3 Retrieval

Query-time audits specify deterministic retrieval with no model calls, on the grounds that retrieval should not itself hallucinate. We implemented exact match plus BM25 over all turns, neighbour expansion, and chronological packing under a 6,000-token cap, then measured whether the packet contained both the turn asserting the old state and the turn asserting the new one.

retrieval	old hit	new hit	pair recall
deterministic lexical	6/25	0/25	0/25
state-indexed	23/25	22/25	20/25

Table 1: Evidence-pair recall, frozen 25-item slice.

Stale supersedes a belief without reusing its vocabulary. The probe asks whether the user still prefers cycling; the superseding turn describes a medical restriction on a wrist. Their content-word overlap is empty. Anchoring retrieval on the query anchors it on the stale state, and the superseding evidence shares little vocabulary with that state.

The new-evidence session always falls in positions 30–39 of 50. We did not exploit that regularity, which fits the benchmark's generator rather than the retrieval problem. A prior selecting only the most recent sessions would also miss the relevant ones, since positions 40–49 are filler.

The index is a one-time per-history pass that tags turns asserting durable user state, yielding a chronological timeline of mean 233 assertions per history (range 168–267). gemini-3.1-flash-lite builds it from the history alone, without access to the probes. Query-time retrieval ranks over that timeline and makes no model calls. The index is byte-identical across arms and content-hash cached, so it cannot bias a between-arm comparison, and we report its cost separately from per-query cost.

4 Capability dependence

With retrieval fixed, we compared audit-and-repair against a no-audit baseline receiving the identical packet, holding everything constant except the generator.

We fixed the selection rule before measuring any effect: take the cheapest candidate whose no-audit accuracy on the development slice reaches 40%. Using only baseline accuracy avoids selecting on the observed contrast, but baseline and treatment outcomes on the same histories are correlated, so selection is not statistically independent of the estimate and regression to the mean remains possible.

candidate	no-audit acc.	$/M in
gemini-3.1-pro-preview	40.0%	2.00
glm-5.3-flash	37.3%	0.075
deepseek-v4-flash	36.0%	0.087
qwen3-235b-a22b	22.7%	0.087

Table 2: Capability screen on the same 25 histories, one no-audit run per candidate with identical retrieval and prompts. Prices are input-token list prices. One candidate cleared the pre-specified 40% threshold.

generator	baseline	repair	gate fires	paired effect (95% CI)
flash-lite, k=5	23%	23%	12–17%	+0.3 $[-2.9, +4.1]$
pro-preview, k=3	41%	55%	25–32%	+14.7 $[+8.7, +20.8]$

Table 3: Identical harness, items, prompts, and verifier; only the generator differs. Accuracy points, n=25 histories.

The weaker generator gives an effect near zero and statistically unresolved (95% CI $[-2.9, +4.1]$; 3 wins, 4 losses, 18 ties); we ran no equivalence test, so this does not establish the absence of a practically relevant effect. The stronger generator gives +14.7 points (p=3×10^{-5}, 95% CI $[+8.7, +20.8]$) with 17 wins and 1 loss. Stale's published overall accuracies for these model classes are 22.4% and 55.2% [Chao et al., 2026], and our baseline arms reproduce that ordering.

The gate accepted a transition on 9–13 of 75 probes across five weaker-generator replicates and on 19–24 of 75 across three stronger-generator replicates. The verifier is deterministic and identical in both conditions, so the difference arises in the four model-mediated stages preceding it: premise decomposition, timeline extraction, the draft check, and transition proposal. Separating them requires stage-level ablations we did not run.

Partial reserve evaluation. Screening and estimation shared the development slice, so a selection effect could inflate +14.7. We therefore evaluated a reserve of 15 histories frozen by the same seed before any experiment and used for no design decision. Budget covered 9 of them at k=1: no-audit 10/27 probes, repair 16/27, paired difference +22.2 points (p=0.016, 95% CI $[+11.1, +33.3]$), 6 wins and 0 losses, with the gate firing on 26% of probes against 25–32% on the development slice. The development estimate lies inside this interval. Because only 9 of 15 reserve histories were run at k=1, this is small, incomplete and imprecise supportive evidence, not a completed held-out protocol; it does not by itself rule out a selection effect. The remaining 6 reserve histories are listed in the supplementary manifest and were never run.

Two generators from one family cannot establish a monotonic relationship between capability and repair benefit. Confirmation across families remains open.

5 Evidence isolation

Prior evaluation of query-time repair, including an earlier run of our own, presents the auditor with only the scenario-relevant sessions. We measured the cost, varying only the packet and recording whether premise decomposition flags a material stale premise.

packet	detection	cited correct turn	tokens
full history, state-indexed	14/25 (56%)	12/14	6,081
pruned timeline	9/25 (36%)	8/9	2,459
oracle, relevant sessions	20/25 (80%)	20/20	3,950

Table 4: Stale-premise detection by packet construction, one run on the same 25 histories with gemini-3.1-pro-preview. Token counts are total prompt length including instructions; the retrieved-evidence budget is capped at 6,000. The design does not isolate which difference between packets produces the gap.

Under isolation the auditor cites the correct superseding turn on every probe where it fires. Among roughly 233 competing state assertions from the same user, detection falls to 56% and citation degrades. The pruned condition detects less than the full packet, so packet length alone does not explain the oracle advantage; information loss, composition and distraction remain confounded. Isolation removes the task of distinguishing one superseding assertion from many plausible non-superseding ones, which is the task a deployed auditor faces. Isolated-evidence accuracy therefore does not represent deployment, and the retrieval condition belongs beside the accuracy in any report.

6 Estimator stability

A subscription endpoint exposing no temperature control gave paired estimates of +26.7 points (p=7e−5) and +9.3 points (p=0.058) from two runs of identical code over identical items, the second outside the first's 95% interval of $[+0.161, +0.408]$. Per history the exact difference agreed on 14 of 25 and the sign on 17 of 25. The two estimates differ by 0.174 against a minimum detectable difference of 0.112.

Temperature 0 on a hosted API did not remove the variation: identical prompts produced varying text and the binary audit decision still flipped on boundary items. Across five replicates the between-run standard deviation was 0.022. That quantity and the 0.174 difference above are not directly comparable, but both bound the run-to-run variation an estimate must survive. This is one pipeline on two endpoints, so we report it as a case study. It does imply that paired memory experiments on hosted inference should report a mean over replicates with its between-run variation rather than a single run with a randomisation p-value.

7 A defect that produced a clean null

An earlier build of our pipeline treated a negative timeline-versus-draft check as a decision to keep the draft, inserting a blocking condition the specification does not authorise: directives derive from premise status, with the deterministic gate as sole authority.

The defective build returned an exactly null estimate with a plausible explanation: detection fired on 11% of probes, so repair could not help. We recorded that explanation before testing it.

build	gate fires	net	paired effect	p
with unauthorised gate	8/75	+0	+0.000	0.58
specification-conformant	21/75	+20	+0.267	7e-5

Table 5: The defective build produced a well-behaved null.

Premise decomposition was in fact firing on 56% of probes, and our own gate was discarding the difference. The diagnostic that caught it measured one pipeline stage in isolation and cost roughly 100 model calls. A defect that suppresses a treatment presents as a null with an available mechanism rather than as a bug, which is why we recommend stage-level instrumentation for pipelines of this shape.

8 What we did not measure

The commit-versus-forget experiment is specified, implemented and unrun. One pass of the two-arm longitudinal matrix costs $70–92 on gemini-3.1-pro-preview against a total project budget of $60. We draw no conclusion about commit versus forget, and none about what the cheaper generator would have shown in that design.

9 Limitations

The judge is a central threat. Every accuracy figure in this paper comes from one LLM judge with no human validation, and on at least one item its label disagreed with the benchmark annotation. Judge error, or judge bias that varies with generator verbosity or style, could change the reported treatment effects. The design called for a second blinded rater over roughly 160 stratified responses; none was secured, so no accuracy figure here should be treated as established, and none appears in the abstract.

The generator comparison is not identified. Replacing the model changes drafting, premise decomposition, timeline extraction, the draft check, transition proposal, and repair simultaneously, so the contrast does not isolate any one stage or capability. Both generators come from a single vendor, so nominal capability is confounded with model version, architecture, and serving behaviour. Factorial stage swaps and cross-vendor models are needed and were not run.

The reimplementation is unvalidated. We did not calibrate against the original authors' reported aggregates or their code, and we found one consequential unauthorised gate in an earlier build of our own pipeline (Section 7). Readers should treat the substrate as a StateAuditor-style pipeline of our construction rather than as a faithful reproduction.

Search protocol. The novelty statement rests on an arXiv and artifact-reachability sweep conducted through 2026-08-11 over memory, staleness, supersession and belief-revision terms. Work that is unpublished, differently described, or released since that date may have been missed.

All results use 25 development histories from one benchmark, one audit reimplementation, one prompt set, and two generators from a single family. Nothing here supports cross-family generality. The capability screen selected its generator at exactly the threshold, and screening and effect estimation used the same histories; the held-out reserve confirms the effect on 9 of 15 reserve histories at k=1, with the remaining 6 unrun for budget reasons rather than excluded. We report no equivalence test for the weaker generator, so its interval does not establish absence of a practically relevant effect. The state index is model-produced, so 20/25 pair recall is a property of the reported system rather than of the data. Stale's probes carry label ambiguity in places, for instance whether a temporary medical restriction supersedes a stated preference or constrains its exercise, and on at least one such item the judge's rubric-based label disagreed with the benchmark annotation. The defect in Section 7 was found in our own code and we do not claim others remain absent.

10 Budget and future work

A hard \$60 inference budget on a single hosted-API account bounded every experiment here, and three results follow directly from it. The two-arm longitudinal commit-versus-forget experiment is specified and implemented but unrun: one pass costs \$70–92 on the only generator that drove the audit. The reserve evaluation covers 9 of 15 frozen histories at one replicate, because a guard halted execution at a fixed credit floor rather than write partial rows. The stronger generator emits roughly 2,000 output tokens per call at \$12 per million output tokens, putting one 25-history gate replicate at \$14, so we ran it three times against five for the weaker generator.

Four extensions would strengthen the claims, in descending order of value per dollar. A factorial stage swap, holding the drafting model fixed while varying the auditing model and the reverse, would isolate whether the observed effect belongs to a particular stage; at cheap-tier prices this is a small number of gate runs. A generator from a different vendor clearing the same threshold would separate capability from model version and vendor, though the three non-Google candidates we screened reached 37.3%, 36.0% and 22.7% against a 40% bar, so this may require a mid-tier model. Completing the frozen reserve with additional replicates would convert a partial evaluation into a completed held-out protocol. The commit-versus-forget experiment itself remains the original question.

Inference budget does not address the largest omission. Every accuracy figure here rests on one LLM judge with no human validation. The design called for a second blinded rater over roughly 160 stratified responses, and securing one is a prerequisite for any accuracy claim in this line of work.

11 Conclusion

The same query-time audit produced a clear gain with the stronger generator and no resolved gain with the weaker one on this development slice, which is consistent with capability dependence without establishing a threshold. Obtaining that comparison required retrieval that reaches superseding evidence in long histories, an evaluation that withholds that evidence from the auditor, and an estimator that survives a second run. Retrieval failure, evidence isolation and endpoint variability each moved the estimated effect by more than the design could resolve.

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